International Journal of Sports Physiology and Performance, 2017, 12, 106-114
http://dx.doi.org/10.1123/ijspp.2015-0743
© 2017 Human Kinetics, Inc.

Human Kinetics
ORIGINAL INVESTIGATION

Single-Leg Power Output and Between-Limbs Imbalances in Team-Sport Players: Unilateral Versus Bilateral Combined Resistance Training

Oliver Gonzalo-Skok, Julio Tous-Fajardo, Luis Suarez-Arrones, José Luis Arjol-Serrano, José Antonio Casajús, and Alberto Mendez-Villanueva

Purpose: To compare the effects of unilateral and bilateral resistance training on single-leg power output, between-limbs imbalance (BLI), bilateral deficit (BLD), change of direction (COD), and linear sprinting and jumping performance in young elite basketball players. ***Methods:*** Twenty-two young (U-16–U-19) male basketball players were randomly assigned either to an exclusive unilateral (UNI) (n = 11) or a bilateral (BIL) (n = 11) resistance-training group during a 6-wk period. Both groups training consisted of 3 unilateral or bilateral 90° back-squat sets. A postdetermined number of repetitions was set until power output dropped to <10% of maximum power (MP) output. In addition, both groups performed 2 sets of 5 unilateral or bilateral drop jumps and 2 sets of 5 unilateral or bilateral countermovement jumps (CMJ). Pre- and posttraining, performance was assessed by an incremental bilateral and unilateral squat-load test, a multiple-COD test (V-cut test), a 15-m-sprint test (7.5 + 7.5 m) with one 180° COD performed with both right (180° RCOD) and left (180° LCOD) legs, a 25-m-sprint test (5- and 15-m split time), and a CMJ test. ***Results:*** Within-group analyses showed substantial improvements in 180° RCOD, bilateral and unilateral MP, 25-m-sprint test, and CMJ in both groups. Between-groups analyses showed substantially better results in 180° LCOD, MP with right and left legs, BLI, and BLD in UNI than in BIL. ***Conclusions:*** Both training programs substantially improved most of the physical-fitness tests, but only UNI reduced between-limbs asymmetry and achieved greater enhancements in actions that mostly required applying force unilaterally in basketball players.

Keywords: asymmetry, change-of-direction ability, instantaneous feedback, plyometrics

The ability to perform high-intensity actions (HIAs) is an important prerequisite for successful participation in most team sports.[1,2] Several studies have reported that HIAs such as acceleration,[3] maximum running speed,[4] change-of-direction (COD) ability,[5] and explosive power[6] can facilitate on-field competitive performance. For example, 83% of goals in soccer are preceded by at least 1 powerful action (eg, linear sprint) of the scoring or the assisting player.[7] Given the importance of HIAs for team-sport performance, training strategies that ensure their optimal development are continuously being investigated.

Specificity is important for training-induced adaptations.[8] Most HIAs in team sports require players to produce force off a single leg when running, bounding, landing, changing direction, or jumping. Moreover, some of the most career-threatening injuries (eg, anterior cruciate ligament [ACL]) typically occur to a 1-legged loaded leg, and when both limbs are loaded the main load is typically on 1 leg (ie, the injured leg).[9,10] In this regard, between-limbs imbalance in strength and power, assessed as the between-limbs imbalance (BLI) or limb-symmetry index, has been considered a valid and useful tool to detect players at high risk (ie, fourfold in players with >10% asymmetry) of lower-extremity injury,[11] as well as to a successful return to sport after an ACL injury.[12] In addition, functional asymmetries might play a role in performance. For example, better symmetrical team-sport players (assessed via unilateral vertical jump or distance reached during a dynamic balance test) seem to be faster than their asymmetrical counterparts during linear- and COD-sprint tests.[13] Therefore, team-sport players might benefit from strength-training interventions aiming to enhance physical performance and/or reduce injury risk that incorporate unilateral strength exercises.

Despite the potential benefits that unilateral strength training might offer to team-sport athletes, very little information is currently available on the topic. For example, it has been shown that the inclusion of unilateral plyometric exercises (ie, combined with bilateral exercises or isolated) was more effective to improve functional performance (ie, jumping, sprinting, COD ability) in young "recreational" soccer players than bilateral plyometric training alone.[14] However, that study employed only plyometric exercises, and single-leg performance (eg, power output and strength) measurements were not conducted. Moreover, to the best of our knowledge, only 1 study has to date investigated the effect of bilateral and unilateral strength training showing similar improvements on physical performance (ie, linear sprint, maximal strength, and COD) in highly trained team-sport players.[15] This scarce information makes it quite difficult to give solid, evidence-based recommendations. Therefore, the aim of the current study was to compare the effects of unilateral and bilateral training on single-leg power output, BLI, bilateral deficit, COD, linear sprinting, and jumping performance in young elite basketball players.

Gonzalo-Skok and Arjol-Serrano are with the Faculty of Health Sciences, University of San Jorge, Zaragoza, Spain. Tous-Fajardo is with the Strength and Conditioning Dept, Italian National Team (FIGC), CTF Coverciano, Italy. Suarez-Arrones is with the Faculty of Sport, Pablo de Olavide University, Seville, Spain. Casajús is with the Growth, Exercise, Nutrition and Development Research Group, Zaragoza, Spain. Mendez-Villanueva is with ASPIRE Academy for Sports Excellence, Doha, Qatar. Address author correspondence to Oliver Gonzalo-Skok at ogonzalo@usj.es.

Downloaded by Ebsco Publishing on 02/08/17, Volume 12, Article Number 1

Methods

Subjects

Twenty-two young (U-16–U-19) highly trained male basketball players (age 16.9 ± 2.1 y, body mass 77.6 ± 9.3 kg, height 189.7 ± 6.9 cm) belonging to an elite basketball club voluntary took part in the current study. Data collection took place in the fourth month of the competitive season after 2 months of the preseason period and 1 month of competition. All players participated on average in ~11 hours of combined basketball (6 sessions) and strength/power (2 sessions) trainings plus 2 competitive matches per week. At the time of the study, all players were competing at national level (ie, Spanish Basketball National League). Furthermore, some players (n = 5) were also competing at international level (ie, European and World Basketball Championship). Every player had a minimum experience of 2 years (range 2–5 y) in strength training. Written informed consent was obtained from both the players and their parents before beginning the investigation. The current study was approved by the institutional research ethics committee and conformed to the recommendations of the Declaration of Helsinki.

Study Design

Using a controlled and randomized study design (ABBA distribution), players were divided into 2 training groups that performed either exclusive unilateral (n = 11; age 16.8 ± 1.7 y, body mass 76.9 ± 8.6 kg, height 190.4 ± 6.9 cm) or bilateral (n = 11; age 16.7 ± 1.7 y, body mass 74.9 ± 9.6 kg, height 188.9 ± 7.5 cm) strength training based on their ranked physical performance. Tests were performed on an indoor basketball court 1 week before training and 1 week after the training period. Tests included a 25-m running sprint test (with 5- and 15-m split times), a countermovement-jump (CMJ) test, a multiple-COD test (V-cut test), a 15-m-sprint test (7.5 m + 7.5 m) with one 180° COD performed both right (180°-CODR) and left (180°-CODL) legs, and an incremental bilateral and unilateral squat-load test. Players were familiarized with the exercise procedures before the commencement of each test. They were asked not to perform intense exercise on the day before a test and to consume their last meal at least 3 hours before the scheduled test time.

Procedures

Training Intervention. Participants performed 2 resistance-training sessions per week, in addition to their normal strength training, for 6 consecutive weeks. The strength-training sessions, performed by both groups, typically included injury-prevention exercises (ie, eccentric strength, balance, and coordination exercises). Unilateral training consisted of 3 unilateral 90°-squat sets (ie, until thighs were parallel to the floor) of a postdetermined number of repetitions. This number was determined by stopping the set when power output (measured via a linear encoder) dropped to <10% of a target power output (maximum power [MP] < 10%) established at the start of the first set in each session. A computer was placed in front of the participants so that they could see the power output in each repetition. After this, players performed 2 sets of 5 unilateral drop jumps (DJs) from a height of 0.25 m and 2 sets of 5 unilateral CMJs, while the only difference in bilateral was the jump height in DJ (0.5 m). Vertical DJs were performed during session 1, whereas horizontal DJs were executed in session 2. Between-sets recovery was 3 minutes in both groups. During unilateral squats, the supporting leg (foot) was placed on an adjustable bench located behind the player. Furthermore, the starting leg in each exercise (ie, squats, DJs, and CMJs) throughout the unilateral training was always the less powerful leg as assessed at the unilateral squat pretest. The concentric phase was performed as fast as possible (with the feet not leaving the floor) and the eccentric phase was slower (ie, self-selected and never exceeding 3 s) in both training groups. The load was progressively increased in the 90°-squat exercise in both groups: 80% of MP load (wk 1), 90% of MP load (wk 2), 100% of MP load (wk 3–5), and 80% of MP load (wk 6). Squat exercise was monitored in each training session through a linear encoder (SmartCoach Power Encoder SPE-35, SmartCoach Europe AB, Stockholm, Sweden).

V-Cut Test. Players performed a 25-m sprint with 4 CODs of 45° each 5 m, which is described elsewhere.[16] The V-cut test was repeated twice, and 3 minutes of passive recovery was provided between repetitions. Time was recorded by timing gates (Microgate, Bolzano, Italy), and the best (fastest) trial was retained. Reliability scores were .92 (90% confidence interval [CI] .87; .95) for intraclass correlation coefficient (ICC) and 1.4% (90%CI 1.2; 1.7%) for coefficient of variation (CV).

180°-COD Test. Players performed a 15-m-sprint test with a 180° COD. Players started from a line, sprinted for 7.5 m, crossed a line with either the right or left foot, and came back to the starting line as fast as possible. Both right (180° RCOD) and left CODs (180° LCOD) were repeated twice. The front foot was placed 0.5 m before the first timing gate. At least 2 minutes of between-repetitions recovery was allowed. Time was recorded by timing gates (Microgate, Bolzano, Italy), and the best time with each 180° COD was retained. Reliability scores were .85 (90%CI .75; .92) and .78 (.62; .87) for ICC and 1.7% (1.4; 2.1%) and 2.1% (1.8; 2.7%) for CV at 180° RCOD and 180° LCOD, respectively.

Incremental Bilateral Squat Test. All squat testing was done in the Smith machine (Salter, Barcelona, Spain) starting with a 20-kg load. Data were registered by a linear encoder (SmartCoach Power Encoder SPE-35, SmartCoach Europe AB, Stockholm, Sweden). Players performed a 90° squat, descending slowly, stopping for approximately 1 second, and ascending at maximal velocity to the initial upright position. The features of the test are described elsewhere.[17] The test ended when the MP with both legs was lower than the previous load.

Incremental Unilateral Squat Test. The procedures were similar to the incremental bilateral test. This test was also performed in a Smith machine (Salter, Barcelona, Spain). The initial leg to be assessed was randomly selected at each load. While performing the unilateral squat in both testing and training, players placed the top of the foot of the nonexecuting leg on a support bar behind them (at 0.8 m height and 1 m of distance from the executing leg) to ensure that the executing leg was the only leg used to perform the squat. If these considerations during the execution were not respected, the repetition was not considered valid. There was a pause between the eccentric and concentric phases. One repetition was performed when velocity in the concentric phase was lower than 1 m/s, whereas 2 repetitions were performed when velocity in the concentric phase was greater than 1 m/s with both the right and left legs with each load. Between-legs recovery was 30 seconds within each load and 3 minutes between loads. The variables retained for analysis were MP with right leg, MP with left leg, and BLI. BLI was based on limb-symmetry index[11]: BLI = MP with worse leg/MP with better leg × 100. Furthermore, bilateral deficit was also analyzed. Bilateral deficit was calculated as follows: {1 – [MP with both limbs/(MP

right leg + MP left leg)]} × 100.[18] ICC was between .88 and .92 and CV was between 3.4% and 5.2% in both unilateral and bilateral MP.

Speed Tests. Running speed was evaluated by 25-m-sprint time (standing start) with 5- and 15-m split times. The front foot was placed 0.5 m before the first timing gate. Time was recorded with photoelectric cells (Microgate, Bolzano, Italy). The 25-m sprint was performed twice, separated by at least 3 minutes of passive recovery. The best time was considered for subsequent analysis. ICC was .73 to .88, while CV was between 1.6% and 2.8%.

Countermovement Jump. Jump height was assessed using vertical CMJ (cm) with flight time measured by an Optojump (Microgate, Bolzano, Italy). Each trial was validated by a visual inspection to ensure that each landing was without any leg flexion, and players were instructed to maintain their hands on their hips during CMJs. The depth of the CMJ was self-selected. Each test was performed 3 times, separated by 45 seconds of passive recovery, and the best jump was recorded. Reliability scores were .94 (90%CI .80; .99) for ICC and 3.3% (2.7; 3.9%) for CV.

Statistical Analyses

Data are presented as mean ± SD. All data were first log-transformed to reduce bias arising from nonuniformity error. The standardized difference or effect size (ES, 90% confidence limit [90%CL]) in the selected variables was calculated using the pooled pretraining SD. Threshold values for Cohen ES statistics were >0.2 (small), >0.6 (moderate), and >1.2 (large).[19] For within/between-groups comparisons, the chances that the differences in performance were better/greater (ie, greater than the smallest worthwhile change, 0.2 × the between-subjects SD, based on Cohen *d* principle), similar, or worse/smaller were calculated. Quantitative chances (QC) of beneficial/better or detrimental/poorer effect were assessed qualitatively as follows: <1%, almost certainly not; 1 to 5%, very unlikely; 5 to 25%, unlikely; 25 to 75%, possible; 75 to 95%, likely; 95 to 99%, very likely; and >99%, almost certain. If the chances of having beneficial/better or detrimental/poorer performances were both >5%, the true difference was assessed as unclear.[19] The Pearson product–moment correlation coefficient was used to determine the relationship between different variables. The following criteria were adopted for interpreting the magnitude of correlation (*r*) between test measures: ≤.1, trivial; >.1 to .3, small; >.3 to .5, moderate; >.5 to .7, large; >.7 to .9, very large; and >.9 to 1.0, almost perfect.[19] If the 90%CL overlapped small positive and negative values, the magnitude of the correlation was deemed unclear; otherwise the magnitude was deemed to be the observed magnitude.[19]

Results

Participants

Only players who participated in >85% of all training sessions were included in the final analyses. Consequently, 4 of the 22 players were excluded for various reasons. None of the players were injured during the strength-training sessions. As a result, 18 players (age 16.8 ± 1.9 y, body mass 77.4 ± 9.2 kg, height 189.7 ± 7 cm) were included in the final analyses. The final sample sizes for the training groups were n = 9 for unilateral and n = 9 for bilateral. In spite of dropouts, no significant differences were found between groups at baseline.

Mean Number of Repetitions and Power Output in the 90°-Squat Exercise

Overall, the bilateral group performed 7.8 ± 0.5 repetitions and 362.8 ± 75.0 W in each set, while the unilateral group executed 8.1 ± 1.3 repetitions and 305.3 ± 35.9 W with right leg and 8.3 ± 1.1 repetitions and 303.7 ± 39.0 W with left leg. The 10% power-output decrement corresponded to a 12.7% ± 0.8% decrement in the mean velocity during all trainings in both groups. The progression throughout relative MP intensity (80%, 90%, and 100%) is illustrated in Figure 1. No substantial differences were found between the mean number of repetitions of any group or leg. Conversely, the bilateral group showed substantially greater mean power output than the unilateral group with right (17.3% [CL90% 0.5; 36.9], ES = 0.86 [0.03; 1.69], QC = 91/7/2%) and with left (15.3% [CL90% 0.9; 27.6], ES = 0.87 [0.05; 1.69], QC = 91/7/2%), while no substantial differences were reported between right and left legs.

Incremental Load Test

MP with both legs was achieved with 47.8 ± 6.8 kg in the bilateral group and 48.3 ± 7.9 kg in the unilateral group. MP with the right leg was achieved with 34.6 ± 8.0 kg in the bilateral group and 36.9 ± 6.6 kg in the unilateral group. MP with the left leg was achieved with 34.3 ± 6.1 kg in the bilateral group and 37.5 ± 5.6 kg in the unilateral group.

Within-Group Changes

Results of within-group changes are presented in Table 1. Substantial improvements were found in 180° RCOD, MP with both legs and left and right leg, all sprint times, and CMJ after both training interventions (unilateral and bilateral). Furthermore, the unilateral group also substantially improved 180° LCOD and BLI while substantially increasing its bilateral deficit.

Between-Groups Changes

Results from between-groups analysis are illustrated in Figure 2. The improvements in 180° LCOD (2.6% [CL90% –0.6; 5.8], QC = 81/15/3%), MP with right leg (12.3% [CL90% –0.7; 26.9], QC = 88/9/3%), MP with left leg (12.8% [CL90% –1.9; 29.8], QC = 84/12/3%), and BLI (53.1% [CL90% 48.0; 350.7], QC = 85/10/5%) were substantially greater in unilateral than in bilateral. Furthermore, a substantially greater bilateral deficit (29.2% [–5.2; 76.0], QC = 83/13/4%) was found in unilateral than in bilateral. A possible performance improvement in CMJ was also found in unilateral compared with bilateral.

Discussion

The aim of the current study was to compare the effects of unilateral and bilateral combined resistance training on squat power output, sprinting, COD, and jumping performance in highly trained young basketball players. The main findings of the study were that postintervention single-leg-squat MP-output (ie, MP with left and right leg) improvements were substantially greater after unilateral than bilateral training; BLIs and bilateral deficit in back-squat MP output were substantially reduced or increased, respectively, only after unilateral training; and small to large posttraining improvements

Downloaded by Ebsco Publishing on 02/08/17, Volume 12, Article Number 1

Downloaded by Ebsco Publishing on 02/08/17, Volume 12, Article Number 1

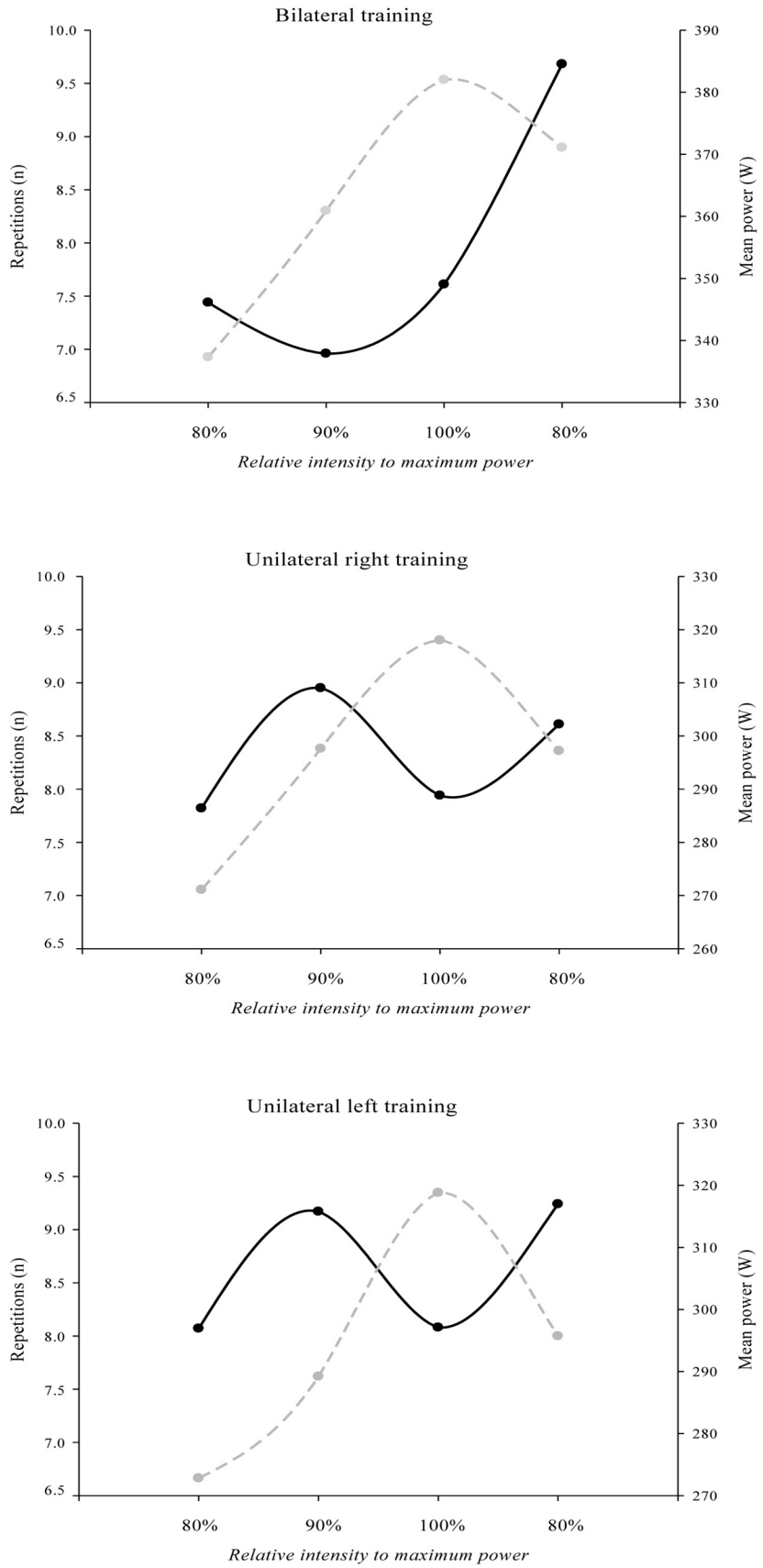

Figure 1 — Mean repetitions and mean power for each training. Solid lines represent mean repetitions and dashed lines represent mean power throughout different relative maximum power intensities (80% wk 1, 90% wk 2, 100% wk 3–5, 80% wk 6).

Downloaded by Ebsco Publishing on 02/08/17, Volume 12, Article Number 1

Table 1 Changes in Athletic Performance After Unilateral Strength Training or Bilateral Strength Training, Mean ± SD

	Unilateral Strength Training (n = 9)					Bilateral Strength Training (n = 9)				
Variable	Pretest	Posttest	Standardized differences (90%CL)	Qualitative assessment	Changes (%, 90%CL)	Pretest	Posttest	Standardized differences (90%CL)	Qualitative assessment	Changes (%, 90% CL)
V-cut test (s)	6.57 ± 0.23	6.50 ± 0.18	0.28 (–0.01; 0.58)	Possibly	1.1 (0.0; 2.2)	6.63 ± 0.24	6.56 ± 0.20	0.26 (–0.28; 0.80)	Unclear	1.0 (–1.2; 3.2)
180° RCOD (s)	3.54 ± 0.15	3.47 ± 0.10	0.43 (0.23; 0.63)	Very likely	2.0 (1.1; 2.9)	3.50 ± 0.12	3.45 ± 0.09	0.41 (–0.05; 0.86)	Likely	1.6 (–0.2; 3.3)
180° LCOD (s)	3.55 ± 0.17	3.46 ± 0.15	0.48 (0.14; 0.82)	Likely	2.6 (0.7; 4.4)	3.48 ± 0.12	3.48 ± 0.11	0.02 (–0.71; 0.74)	Unclear	0.1 (–2.7; 2.8)
MPB (W)	406.9 ± 48.5	466.3 ± 72.8	0.93 (0.70; 1.16)	Almost certainly	14.2 (10.5; 18.0)	407.5 ± 56.2	470.2 ± 75.8	0.89 (0.52; 1.26)	Almost certainly	15.0 (8.5; 21.9)
MPR (W)	273.7 ± 40.0	372.2 ± 57.3	1.74 (1.28; 2.20)	Almost certainly	35.8 (25.3; 47.2)	270.1 ± 32.8	330.1 ± 61.2	1.38 (0.63; 2.13)	Very likely	21.0 (9.1; 34.2)
MPL (W)	271.2 ± 36.8	370.4 ± 50.4	1.90 (1.25; 2.56)	Almost certainly	36.7 (22.8; 52.1)	270.9 ± 51.3	330.4 ± 73.8	0.83 (0.37; 1.29)	Very likely	21.1 (8.9; 34.7)
BLI (%)	9.6 ± 3.8	4.8 ± 1.3	1.15 (0.42; 1.87)	Very likely	46.5 (20.4; 64.0)	6.9 ± 5.0	4.4 ± 2.5	0.40 (–0.49; 1.30)	Unclear	32.6 (–62.1; 72.0)
BLD (%)	24.8 ± 8.9	36.8 ± 7.7	1.00 (0.52; 1.49)	Likely	55.6 (25.7; 92.7)	23.1 ± 8.4	27.0 ± 6.4	0.41 (–0.23; 1.04)	Unclear	20.5 (–6.8; 55.7)
5-m (s)	1.13 ± 0.05	1.07 ± 0.06	0.92 (0.51; 1.33)	Almost certainly	5.0 (2.8; 7.1)	1.10 ± 0.05	1.06 ± 0.02	0.72 (0.23; 1.22)	Very likely	3.7 (1.3; 6.0)
15-m (s)	2.54 ± 0.08	2.48 ± 0.1	0.76 (0.51; 1.02)	Almost certainly	2.7 (1.8; 3.6)	2.52 ± 0.09	2.46 ± 0.07	0.55 (0.31; 0.79)	Very likely	2.1 (1.2; 2.9)
25-m (s)	3.84 ± 0.12	3.75 ± 0.15	0.70 (0.39; 1.01)	Almost certainly	2.4 (1.3; 3.4)	3.80 ± 0.15	3.69 ± 0.15	0.70 (0.36; 1.03)	Very likely	2.9 (1.5; 4.3)
CMJ (cm)	37.4 ± 4.2	39.8 ± 5.1	0.47 (0.27; 0.67)	Very likely	6.5 (3.7; 9.3)	38.9 ± 5.3	40.6 ± 5.5	0.27 (0.15; 0.38)	Likely	4.5 (2.5; 6.4)

Abbreviations: V-cut test, 25-m sprint with 4 changes of direction; 180° RCOD, 15-m (7.5 + 7.5) sprint with 1 change of direction of 180° performed with right leg; 180° LCOD, 15-m (7.5 + 7.5) sprint with 1 change of direction of 180° performed with left leg; MPB, maximum power exerted during bilateral squat; MPR, maximum power with right leg exerted during unilateral squat; MPL, maximum power with left leg exerted during unilateral squat; BLI, between-limbs imbalance; BLD, bilateral deficit; 5-m, 5-m sprint; 15-m, 15-m sprint; 25-m, 25-m sprint; CMJ, countermovement jump; CL, confidence limits. Note: standardized differences = effect size.

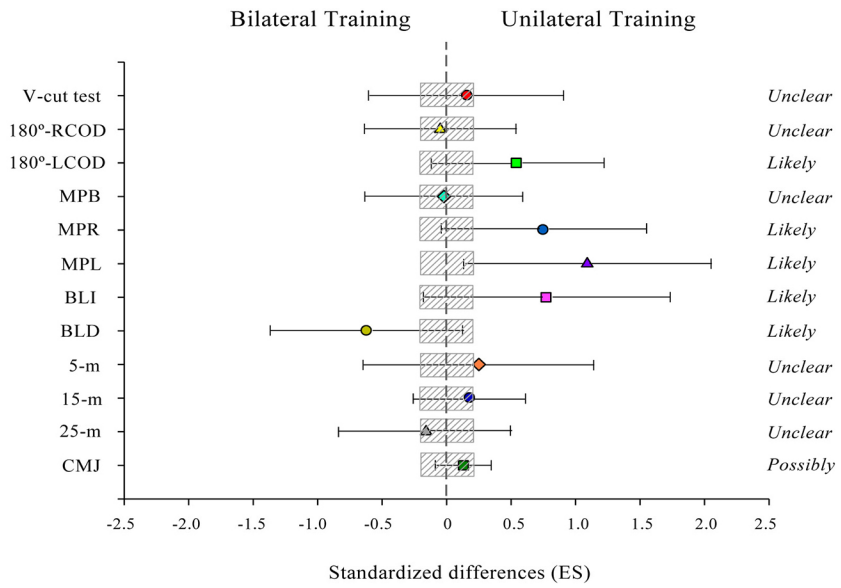

Figure 2 — Efficiency of the unilateral compared with bilateral strength-training program to improve multiple changes of direction (V-cut test); 180° change of direction with left (180°-LCOD) and right (180°-RCOD) legs; maximum power bilaterally (MPB); maximum power with right leg (MPR) and with left leg (MPL); between-limbs imbalance (BLI); bilateral deficit (bilateral deficit); 5-, 15-, and 25-m sprint time; and countermovement-jump performance (CMJ) (bars indicate uncertainty in the true mean changes with 90% confidence limits). Trivial areas were the smallest worthwhile change (see Methods).

Downloaded by Ebsco Publishing on 02/08/17, Volume 12, Article Number 1

in power output, sprinting, and jumping performance were found in both groups.

Most game-related HIAs in team sports, such as accelerating, braking, changing direction, jumping, landing, or kicking, typically occur with both legs loaded asymmetrically. These mainly unilateral actions can also be linked to some of the most common (eg, ankle and knee sprains) and serious (eg, ACL) injuries suffered by team-sport players.[9,10] Therefore, within this cohort, the improvement of single-leg function (eg, unilateral power or strength) may be important for both on-field physical performance and protection against injury. In the current study, posttraining improvements in single-leg power output (ie, MP with left and right legs) were substantially greater in unilateral than in bilateral (Table 1 and Figure 1). While to our knowledge this is the first time that power output has been assessed after unilateral training with a linear encoder (previously a Margaria-Kalamen stair-climb test or unilateral vertical-jumping power have been used[20]), present results are somehow at odds with what was reported in a recent study with rugby players.[15] Similar absolute strength unilateral and bilateral improvements (ie, 1RM) were shown after both bilateral (ie, back squat) (ES = 1.36) and unilateral (ie, elevated split squat) (ES = 1.25) resistance training.[15] In addition to the different metrics used (MP output vs 1RM), between-studies differences in the training-load prescription (instantaneous feedback with individualized number of repetitions vs predefined number of repetitions), training orientation (power vs maximal loads), or exercise selection (strength training and plyometrics vs strength training) might be behind the somehow contrasting findings. To our knowledge, there are no other published studies examining the effect of unilateral or bilateral strength training on single-leg strength and/or power in team-sport players.

Another interesting finding of the current study was the substantial reduction in BLI observed after unilateral training, which was likely greater than in the bilateral group. Despite the apparent critical relevance of BLI for both protection against injury and performance,[11,13] to the best of our knowledge, this is the first study to assess the impact of a training intervention on BLI in team-sport players. Therefore, comparisons are not possible. A previous study evaluated the effect of unilateral balance/strength exercises in comparison with a control group in young tennis players.[21] Note that a substantial decrease in BLI in a single-leg horizontal (ES = 2.07) and lateral hop (ES = 2.08) was found in the experimental group, while the control group maintained BLI values (ES = –0.06 to –0.02). Therefore, despite the fact that more research is needed, it seems that training interventions involving unilateral strength and balance/motor-control actions might prove effective in reducing BLI.

In a similar way, bilateral deficit was largely increased only after unilateral training, with likely greater changes than the bilateral group. Considering the observed larger improvements in single-leg power output in comparison with 2-leg squat in the unilateral group, these findings were expected.[22,23] In this regard, changes in unilateral-squat power output were strongly correlated with changes in bilateral deficit (0.85 [CL90% 0.64; 0.94]). On the contrary, the large effect of bilateral training on unilateral power and its negative effect on bilateral deficit was somewhat unexpected. It is uncertain whether the habitual strength-training sessions or the addition of the unique combined training program originated these adaptations, but the latter is an intriguing possibility that clearly deserves further research.

COD ability is believed to depend on a myriad of factors such as technique, linear sprinting speed, leg-muscle qualities, and anthropometry.[24] As such, the improvement of leg-muscle qualities (eg, reactive strength, concentric strength and power, and BLI) might aid in enhancing COD ability. Our results showed that unilateral training induced some small improvements in COD ability in the 3 tests conducted, while only 1 test was improved after the bilateral training (Table 1). The observed magnitude of

Downloaded by Ebsco Publishing on 02/08/17, Volume 12, Article Number 1

the substantial changes (ie, improvements) in COD ability was lower (ES = 0.28–0.48) than what has been previously reported in late-adolescent team-sport players after either bilateral or unilateral strength training (ES = 0.48–1.86).[15,25] Others have reported no changes (ES = –2.0 to 0.3) in COD ability after a complex-contrast training or a combined conditioning training (strength + aerobic high intensity + specific rugby skills) in highly trained team-sport players.[26–28] Between-studies outcome differences might be due to different training approaches employed, such as training volume (7 sets in the current study vs 14–18 sets[25] or 44 vs 82–100 repetitions per training session), players' resistance-training experience (2 vs 1 y[15]), training intensity (near or equal to MP loads and plyometrics vs 75–92% 1RM[15]), or the moment of the competitive season (preseason vs in-season). Bilateral training was never more effective than unilateral at improving COD performance (Figure 1). Somehow similarly, previous studies comparing unilateral with bilateral training have shown substantially greater improvements after unilateral than bilateral,[15,25] supporting the idea of the principle of specificity.[8] In the current study, unilateral training was more effective than bilateral at enhancing left-side single-COD performance (180° LCOD), whereas an unclear outcome was provided in 180° RCOD (Figure 1). The fact that unilateral training was superior to bilateral in improving only the ability to turn to the left side might be related to the specific overload of that (weak) side imposed only during unilateral training. It has been reported that during bilateral back-squat training, vertical-ground-reaction-force asymmetry is significantly different between limbs (4.3%) at the beginning of a set.[29] Furthermore, this magnitude given by percentage difference between left and right legs in vertical ground-reaction force favoring the preferred leg is maintained at the end of a set in asymmetric subjects.[29] In this regard, our basketball players were all right-leg dominant (determined from the self-reported preferred leg via a questionnaire) and were considered asymmetric participants. Hence, these differences could be because players who only train bilaterally might exert much effort with right leg during the whole set, while players training unilaterally were systematically exposed to left-leg efforts.

The strength-training approach employed in the current study (ie, combined training compounded by resistance training + plyometrics), rather than isolated traditional strength-training methods, is considered as an optimal training strategy to enhance sprint performance.[30] As such, substantially better linear-sprinting performance was found after the training period in both groups (ie, unilateral and bilateral). These results (ES = 0.55–0.92) are within the range of or even greater than (ES = –0.1 to 0.87) those of previous training studies after unilateral or bilateral resistance training in male team-sport players.[15,25] In the current study, no substantial differences in sprinting performance between unilateral and bilateral training interventions were found. Previous studies on the impact of unilateral and bilateral strength training on sprinting performance reported contrasting findings.[15,25] One of those studies reported substantially better sprinting performance after unilateral training in comparison with bilateral (ES = 0.24 vs 0.58),[15] whereas a greater bilateral training effect (ES = –0.10 vs 0.70) was found in the other study.[25] Differences in the team-sport players investigated (college rugby vs academy rugby vs basketball), sprinting distance assessment (10 vs 25 vs 40 m), and the training performed (resistance training + plyometrics with individualized repetitions number vs resistance training with predefined repetitions vs combined training) might explain these between-studies differences. Further research is warranted to better understand the effect of unilateral or bilateral resistance training on linear-sprinting performance.

The present results indicated that 6 weeks of combined strength training had a substantial beneficial impact on jumping performance (ES unilateral = 0.47, ES bilateral = 0.27). Despite jumping performance's being considered an important physical-fitness component in most team sports, this is the first study that investigated the effect of unilateral resistance training on CMJ performance in highly trained team-sport players. Therefore, direct comparisons are not possible. The CMJ changes found in the current study after unilateral training (ES = 0.47) are within the range (ES = 0.19–0.79) of previously published reports after different bilateral training strategies in young male team-sport players.[27,31–33] In this regard, interventions including bilateral resistance training, complex training, and explosive strength training have typically reported greater training effects (ES = 0.53, 0.62, 0.78)[23,27–29] than plyometric training (ES = 0.33)[32] and complex-contrast strength training (ES = 0.19).[27] It should be noted that all studies that found greater training-induced changes in CMJ than in the current study were performed with younger team-sport players (14 vs 17 y) or those more inexperienced in strength training (novices vs 2 y experience). In this regard, it is well known that age and training background can influence the magnitude of training adaptation.[6] Moreover, basketball players are typically required to perform many more jumps during matches and practices than soccer players. As such, it might be possible that this population (basketball players) would have more difficulties in improving vertical-jumping performance.

A unique feature of the current study was that both training groups performed a postdetermined number of repetitions of the 90°-squat exercise based on a minimum accepted target power output. While this could be seen as a potential limitation for nonequivalent training volumes, the overall number of repetitions in each condition throughout the training program was similar. However, we cannot know whether the target power-output challenge and/or the provision of instantaneous feedback may have affected the positive results since the conditions were the same for both training groups. It is expected that an unknown training duration (repetitions per set) may have implications on anticipatory/feedforward components (pacing strategies) by means of adjustments in power output.[34] As for feedback provision, even though the literature concerning its long-term effectiveness is scarce, it is believed to be a useful tool to individualize and optimize training interventions by providing a more consistent performance while avoiding repetitions that might produce a negative effect on neuromuscular performance.[35]

There are some limitations of this study. It is uncertain whether performance was enhanced through the effect of individualized MP training, the combination of strength and plyometrics, or both training strategies. As such, further studies are warranted to analyze the influence of performing a predetermined versus postdetermined number of repetitions based on a minimum target power and different combinations of resistance-training methods. In addition, more functional tests such as unilateral jumps in different axes or the Star Excursion Balance Test to assess dynamic stability should be included.

Conclusions

Combined unilateral training reduced between-limbs differences (ie, asymmetry) and achieved greater enhancements in actions that require applying force unilaterally than did bilateral training.

In addition, unilateral training induced substantial adaptations in jumping, linear sprinting, and COD ability.

Practical Applications

Strength training is typically prescribed bilaterally, although the most important actions in team sports (ie, jumps, sprints, COD) are performed mainly unilaterally. As such, based on the present results, the inclusion of unilateral strength-training exercises can be of interest for players with between-legs strength and power asymmetries, as well for players who want to enhance their ability to produce force unilaterally. Finally, the use of a target power-output challenge by providing instantaneous feedback might be seen as an effective tool not only to optimize strength training but also to motivate athletes.

References

1. Ben Abdelkrim N, El Fazaa S, El Ati J. Time-motion analysis and physiological data of elite under-19-year-old basketball players during competition. *Br J Sports Med*. 2007;41:69–75, discussion 75. PubMed doi:10.1136/bjsm.2006.032318
2. Gabbett T, King T, Jenkins D. Applied physiology of rugby league. *Sports Med*. 2008;38:119–138. PubMed doi:10.2165/00007256-200838020-00003
3. Chaouachi A, Brughelli M, Chamari K, et al. Lower limb maximal dynamic strength and agility determinants in elite basketball players. *J Strength Cond Res*. 2009;23:1570–1577. PubMed doi:10.1519/JSC.0b013e3181a4e7f0
4. Ben Abdelkrim N, Chaouachi A, Chamari K, Chtara M, Castagna C. Positional role and competitive-level differences in elite-level men's basketball players. *J Strength Cond Res*. 2010;24:1346–1355. PubMed doi:10.1519/JSC.0b013e3181cf7510
5. Ben Abdelkrim N, Castagna C, Jabri I, Battikh T, El Fazaa S, El Ati J. Activity profile and physiological requirements of junior elite basketball players in relation to aerobic-anaerobic fitness. *J Strength Cond Res*. 2010;24:2330–2342. PubMed doi:10.1519/JSC.0b013e3181e381c1
6. Ziv G, Lidor R. Vertical jump in female and male basketball players—a review of observational and experimental studies. *J Sci Med Sport*. 2010;13:332–339. PubMed doi:10.1016/j.jsams.2009.02.009
7. Faude O, Koch T, Meyer T. Straight sprinting is the most frequent action in goal situations in professional football. *J Sports Sci*. 2012;30:625–631. PubMed doi:10.1080/02640414.2012.665940
8. Sáez de Villarreal E, Requena B, Cronin JB. The effects of plyometric training on sprint performance: a meta-analysis. *J Strength Cond Res*. 2012;26:575–584. PubMed doi:10.1519/JSC.0b013e318220fd03
9. Waldén M, Krosshaug T, Bjorneboe J, Andersen TE, Faul O, Hagglund M. Three distinct mechanisms predominate in noncontact anterior cruciate ligament injuries in male professional football players: a systematic video analysis of 39 cases. *Br J Sports Med*. 2015;49:1452–1460. PubMed doi:10.1136/bjsports-2014-094573
10. Koga H, Nakamae A, Shima Y, et al. Mechanisms for noncontact anterior cruciate ligament injuries: knee joint kinematics in 10 injury situations from female team handball and basketball. *Am J Sports Med*. 2010;38:2218–2225. PubMed doi:10.1177/0363546510373570
11. Gustavsson A, Neeter C, Thomee P, et al. A test battery for evaluating hop performance in patients with an ACL injury and patients who have undergone ACL reconstruction. *Knee Surg Sports Traumatol Arthrosc*. 2006;14:778–788. PubMed doi:10.1007/s00167-006-0045-6
12. Ardern CL, Webster KE, Taylor NF, Feller JA. Return to the pre-injury level of competitive sport after anterior cruciate ligament reconstruction surgery: two-thirds of patients have not returned by 12 months after surgery. *Am J Sports Med*. 2011;39:538–543. PubMed doi:10.1177/0363546510384798
13. Lockie RG, Callaghan SJ, Berry SP, et al. Relationship between unilateral jumping ability and asymmetry on multidirectional speed in team-sport athletes. *J Strength Cond Res*. 2014;28:3557–3566. PubMed doi:10.1519/JSC.0000000000000588
14. Ramírez-Campillo R, Burgos C, Henriquez-Olguin C, et al. Effect of unilateral, bilateral and combined plyometric training on explosive and endurance performance of young soccer players. *J Strength Cond Res*. 2015;29:1317–1328. PubMed doi:10.1519/JSC.0000000000000762
15. Speirs DE, Bennett M, Finn CV, Turner AP. Unilateral vs bilateral squat training for strength, sprints and agility in academy rugby players. *J Strength Cond Res*. 2016;30:386–392. PubMed doi:10.1519/JSC.0000000000001096
16. Gonzalo-Skok O, Tous-Fajardo J, Suarez-Arrones L, Arjol-Serrano JL, Casajus JA, Mendez-Villanueva A. Validity of the V-cut Test for young basketball players. *Int J Sports Med*. 2015;36:893–899. PubMed doi:10.1055/s-0035-1554635
17. de Hoyo M, Gonzalo-Skok O, Sañudo B, et al. Comparative effects of in-season full back squat, resisted sprint and plyometric training on explosive performance in U-19 elite soccer players. *J Strength Cond Res*. 2016;30:368–377. PubMed doi:10.1519/JSC.0000000000001094
18. Rejc E, Lazzer S, Antonutto G, Isola M, di Prampero PE. Bilateral deficit and EMG activity during explosive lower limb contractions against different overloads. *Eur J Appl Physiol*. 2010;108:157–165. PubMed doi:10.1007/s00421-009-1199-y
19. Hopkins WG, Marshall SW, Batterham AM, Hanin J. Progressive statistics for studies in sports medicine and exercise science. *Med Sci Sports Exerc*. 2009;41:3–13. PubMed doi:10.1249/MSS.0b013e31818cb278
20. McCurdy KW, Langford GA, Doscher MW, Wiley LP, Mallard KG. The effects of short-term unilateral and bilateral lower-body resistance training on measures of strength and power. *J Strength Cond Res*. 2005;19:9–15. PubMed
21. Sannicandro I, Cofano G, Rosa RA, Piccinno A. Balance training exercises decrease lower-limb strength asymmetry in young tennis players. *J Sports Sci Med*. 2014;13:397–402. PubMed
22. Taniguchi Y. Lateral specificity in resistance training: the effect of bilateral and unilateral training. *Eur J Appl Physiol Occup Physiol*. 1997;75:144–150. PubMed doi:10.1007/s004210050139
23. Taniguchi Y. Relationship between the modifications of bilateral deficit in upper and lower limbs by resistance training in humans. *Eur J Appl Physiol Occup Physiol*. 1998;78:226–230. PubMed doi:10.1007/s004210050411
24. Sheppard JM, Young WB. Agility literature review: classifications, training and testing. *J Sports Sci*. 2006;24:919–932. PubMed doi:10.1080/02640410500457109
25. Fisher J, Wallin M. Unilateral versus bilateral lower-body resistance and plyometric training for change of direction speed. *J Athl Enhanc*. 2014;3:1–5.
26. Mujika I, Santisteban J, Castagna C. In-season effect of short-term sprint and power training programs on elite junior soccer players. *J Strength Cond Res*. 2009;23:2581–2587. PubMed doi:10.1519/JSC.0b013e3181bc1aac

Downloaded by Ebsco Publishing on 02/08/17, Volume 12, Article Number 1

27. Maio Alves JM, Rebelo AN, Abrantes C, Sampaio J. Short-term effects of complex and contrast training in soccer players' vertical jump, sprint, and agility abilities. *J Strength Cond Res*. 2010;24:936–941. PubMed doi:10.1519/JSC.0b013e3181c7c5fd
28. Gabbett TJ, Johns J, Riemann M. Performance changes following training in junior rugby league players. *J Strength Cond Res*. 2008;22:910–917. PubMed doi:10.1519/JSC.0b013e31816a5fa5
29. Hodges SJ, Patrick RJ, Reiser RF, 2nd. Effects of fatigue on bilateral ground reaction force asymmetries during the squat exercise. *J Strength Cond Res*. 2011;25:3107–3117. PubMed doi:10.1519/JSC.0b013e318212de7b
30. Seitz LB, Reyes A, Tran TT, de Villarreal ES, Haff GG. Increases in lower-body strength transfer positively to sprint performance: a systematic review with meta-analysis. *Sports Med*. 2014;44:1693–1702. PubMed doi:10.1007/s40279-014-0227-1
31. Santos EJ, Janeira MA. Effects of complex training on explosive strength in adolescent male basketball players. *J Strength Cond Res*. 2008;22:903–909. PubMed doi:10.1519/JSC.0b013e31816a59f2
32. Chelly MS, Ghenem MA, Abid K, Hermassi S, Tabka Z, Shephard RJ. Effects of in-season short-term plyometric training program on leg power, jump- and sprint performance of soccer players. *J Strength Cond Res*. 2010;24:2670–2676. PubMed doi:10.1519/JSC.0b013e3181e2728f
33. Buchheit M, Mendez-Villanueva A, Delhomel G, Brughelli M, Ahmaidi S. Improving repeated sprint ability in young elite soccer players: repeated shuttle sprints vs. explosive strength training. *J Strength Cond Res*. 2010;24:2715–2722. PubMed doi:10.1519/JSC.0b013e3181bf0223
34. Tucker R. The anticipatory regulation of performance: the physiological basis for pacing strategies and the development of a perception-based model for exercise performance. *Br J Sports Med*. 2009;43:392–400. PubMed doi:10.1136/bjsm.2008.050799
35. Randell AD, Cronin JB, Keogh JW, Gill ND, Pedersen MC. Effect of instantaneous performance feedback during 6 weeks of velocity-based resistance training on sport-specific performance tests. *J Strength Cond Res*. 2011;25:87–93. PubMed doi:10.1519/JSC.0b013e3181fee634

Downloaded by Ebsco Publishing on 02/08/17, Volume 12, Article Number 1

Copyright of International Journal of Sports Physiology & Performance is the property of Human Kinetics Publishers, Inc. and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.